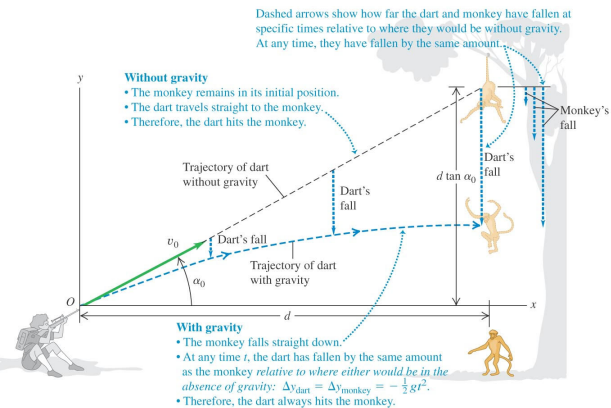


Lecture 2 - Position, velocity, and acceleration of a particle in 2D

- A/Prof Tay Seng Chuan
Email: pvotaysc@nus.edu.sg

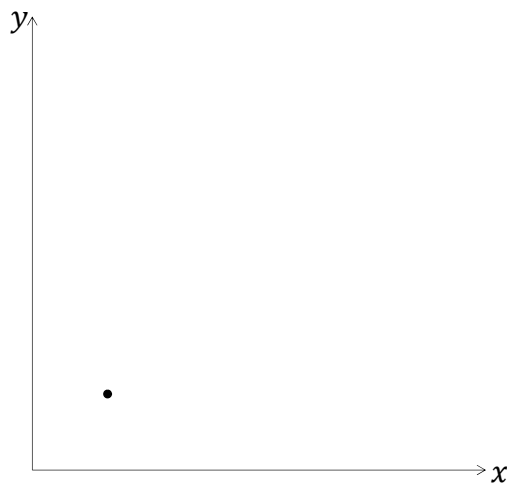
In this lecture we consider the motion of a particle in two dimensions. A few slides in this chapter come from Chapter-3 of Young's textbook.



1

Position, velocity, and acceleration of a particle on (x, y) plane

Consider a particle of mass m , moving on the xy -plane:



2

Position, velocity, and acceleration of a particle on (x, y) plane (cont.)

The **position** of the particle, (x, y) , changes with **time** t .

x and y are **functions** of time t , i. e., $x = x(t)$, and $y = y(t)$.

The **velocity** of the particle is represented by (v_x, v_y) with

$$v_x = v_x(t) = \frac{d}{dt}(x(t)), \text{ and } v_y = v_y(t) = \frac{d}{dt}(y(t)).$$

The **speed**, or magnitude of the **velocity**, of the particle is $v = \sqrt{v_x^2 + v_y^2}$.

The **acceleration** of the particle is represented by (a_x, a_y) with

$$a_x = a_x(t) = \frac{d}{dt}(v_x(t)) = \frac{d^2}{dt^2}(x(t)), \text{ and}$$

$$a_y = a_y(t) = \frac{d}{dt}(v_y(t)) = \frac{d^2}{dt^2}(y(t)).$$

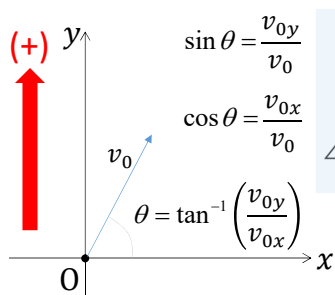
The magnitude of the **acceleration** of the particle is $a = \sqrt{a_x^2 + a_y^2}$.

3

Example 1. Projectile motion ($a_x = 0, a_y = -g$)

$$x(t) = x_0 + v_{0x}t, \quad y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

where $g = 9.80 \text{ m/s}^2$.



At time $t = 0$, the **position** of the particle is $(x(0), y(0)) = (x_0, y_0)$.

The **velocity** of the particle,

$$v_x(t) = \frac{d}{dt}(x(t)) = v_{0x}$$

$$v_y(t) = \frac{d}{dt}(y(t)) = v_{0y} - gt$$

$$\vec{v}(t) = \vec{v}_x(t) + \vec{v}_y(t)$$

At time $t = 0$, the **velocity** of the particle is $(v_x(0), v_y(0)) = (v_{0x}, v_{0y})$ and $\theta = \tan^{-1}\left(\frac{v_{0y}}{v_{0x}}\right)$.

The **acceleration** of the particle is

$$a_x(t) = \frac{d}{dt}(v_x(t)) = 0$$

$$a_y(t) = \frac{d}{dt}(v_y(t)) = -g \text{ Why -ve?}$$

4

Example 1. Projectile motion (cont.)

Remark 1.

$$x = x(t) = x_0 + v_{0x}t \rightarrow t = \frac{x - x_0}{v_{0x}}$$

$$y = y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

Substituting $t = \frac{x - x_0}{v_{0x}}$ into $y(t)$, we have

$$y = y_0 + v_{0y}\left(\frac{x - x_0}{v_{0x}}\right) - \frac{1}{2}g\left(\frac{x - x_0}{v_{0x}}\right)^2$$

$$\rightarrow y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0$$

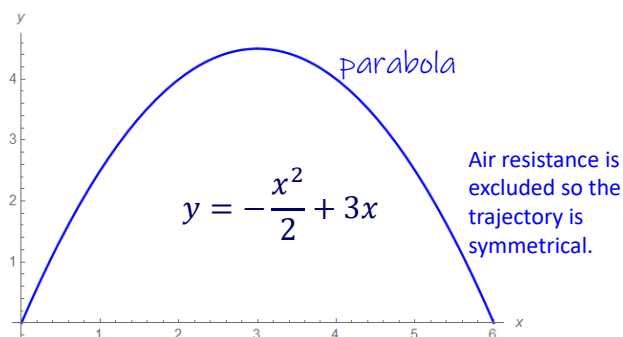
5

Example 1. Projectile motion (cont.)

If $x_0 = y_0 = 0$, $g = 9.8$, $v_{0x} = \sqrt{9.8}$, and $v_{0y} = 3v_{0x}$

$$y = -\frac{g}{2v_{0x}^2}(x - x_0)^2 + \frac{v_{0y}}{v_{0x}}(x - x_0) + y_0$$

$$= -\frac{g}{2(\sqrt{9.8})^2}(x - 0)^2 + \frac{3v_{0x}}{v_{0x}}(x - 0) + 0 \rightarrow y = -\frac{x^2}{2} + 3x$$



6

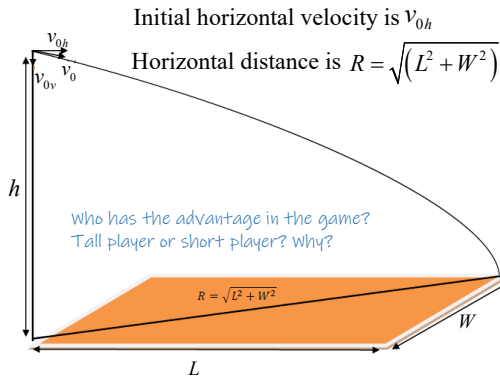
Problem. A tennis player wants to slam a serve from one corner of the court at height h above the ground so that the ball lands just inside the opposite corner of the court.

The court has length L and width W . What should the ratio $\frac{v_{0v}}{v_{0h}}$ be, where v_{0v} is the vertical component, and v_{0h} is the horizontal component of the initial velocity v_0 ?

The time, $t = 0$ is the time when the ball is hit by the racket, and $t = \tau$ is the time when the ball touches the ground.

Answer:

Projectile motion in horizontal direction:



Total time taken is τ

$$v_{0h} = \frac{\text{distance}}{\text{time}} = \frac{R}{\tau} = \frac{\sqrt{L^2 + W^2}}{\tau} \dots (1)$$

Projectile motion in vertical plane:

Initial vertical downward velocity is v_{0v} .

Vertical downward distance is $y = h$.

Total time taken is τ .

$$h = v_{0v} \times \tau + \frac{1}{2} g \tau^2$$

$$\Rightarrow v_{0v} = \frac{h - \frac{g\tau^2}{2}}{\tau} \dots (2)$$

$$\begin{aligned} \frac{(2)}{(1)}: \frac{v_{0v}}{v_{0h}} &= \frac{\left(\frac{h - \frac{g\tau^2}{2}}{\tau} \right)}{\left(\frac{\sqrt{L^2 + W^2}}{\tau} \right)} \\ &= \frac{h - \frac{g\tau^2}{2}}{\sqrt{L^2 + W^2}} \end{aligned}$$

7

Problem. A particle is projected with a speed of 20 m/s. At a distance of 10 m from the point of projection, there is a vertical wall of height 10 m. This wall has a hole of height 3 m, and the base of the hole is 5 m above the horizontal plane through the point of projection. Find the range(s) of the angle of projection which will enable the particle to pass through this hole.

Answer:

$$\text{From } \vec{v}(t) = \vec{v}_x(t) + \vec{v}_y(t) \Rightarrow \cos \theta = \frac{v_{0x}}{v_0} \Rightarrow v_{0x} = v_0 \times \cos \theta.$$

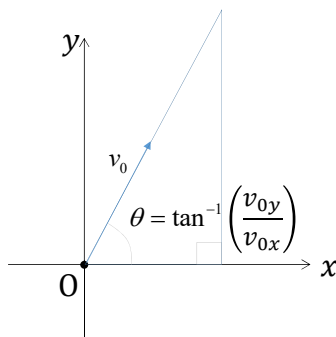
Let $x_0 = 0$ and $y_0 = 0$ in

$$y = -\frac{g}{2v_{0x}^2} (x - x_0)^2 + \frac{v_{0y}}{v_{0x}} (x - x_0) + y_0 \dots (1)$$

(1) becomes

$$\begin{aligned} y &= -\frac{g}{2v_0^2} x^2 + (\tan \theta)x + 0 = -\frac{gx^2}{2v_0^2 \cos^2 \theta} + (\tan \theta)x + 0 \\ &= -\frac{gx^2}{2v_0^2} (\sec^2 \theta) + (\tan \theta)x = -\frac{gx^2}{2v_0^2} (1 + \tan^2 \theta) + (\tan \theta)x \end{aligned}$$

$$\Rightarrow y = x \tan \theta - \frac{gx^2}{2v_0^2} (1 + \tan^2 \theta) \dots (2)$$



8

Continue ... A particle is projected with a speed of 20 m/s. At a distance of 10 m from the point of projection, there is a vertical wall of height 10 m. This wall has a hole of height 3 m, and the base of the hole is 5 m above the horizontal plane through the point of projection. Find the range(s) of the angle of projection which will enable the particle to pass through this hole.

$$y = x \tan \theta - \frac{gx^2}{2v_0^2} (1 + \tan^2 \theta) \dots (2)$$

Lock in (10, 5) and $v_0 = 20$ m/s to (2):

$$5 = 10 \tan \theta - \frac{9.8(10)^2}{2(20)^2} (1 + \tan^2 \theta)$$

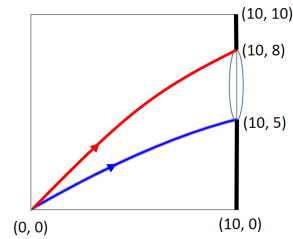
$$\rightarrow 5 = 10 \tan \theta - 1.225 (1 + \tan^2 \theta)$$

$$1.225 \tan^2 \theta - 10 \tan \theta + 6.225 = 0$$

$$\tan \theta = \frac{10 \pm \sqrt{(-10)^2 - 4(1.225)(6.225)}}{2(1.225)}$$

$$= \frac{10 \pm 8.3365}{2.45} = 7.4843, 0.6789$$

$$\theta = 82.38^\circ, 34.16^\circ$$



Lock in (10, 8) and $v_0 = 20$ m/s to (2):

$$8 = 10 \tan \theta - 1.225 (1 + \tan^2 \theta)$$

$$\rightarrow 1.225 \tan^2 \theta - 10 \tan \theta - 9.225 = 0$$

$$\tan \theta = \frac{10 \pm \sqrt{(-10)^2 - 4(1.225)(-9.225)}}{2(1.225)}$$

$$= \frac{10 \pm 7.4025}{2.45} = 7.1031, 1.0602$$

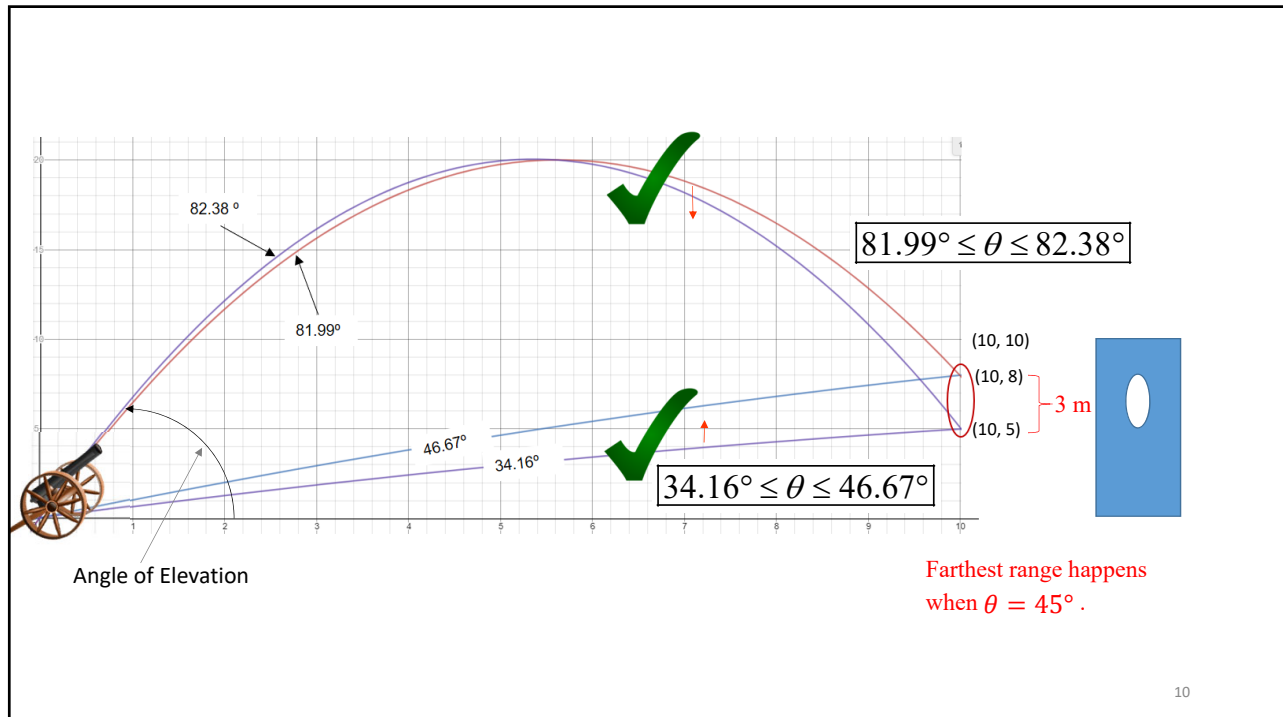
$$\theta = 81.99^\circ, 46.67^\circ$$

Thus the ranges of θ are $34.16^\circ \leq \theta \leq 46.67^\circ$

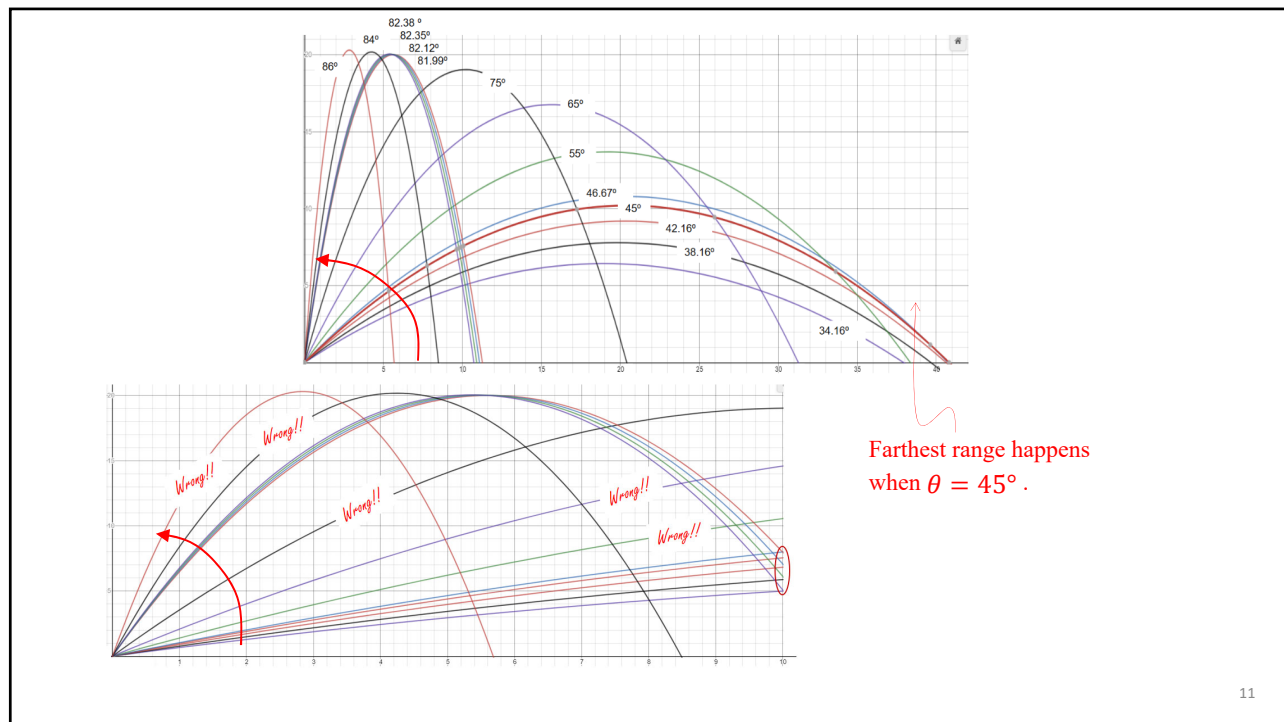
and $81.99^\circ \leq \theta \leq 82.38^\circ$

Why ??

9

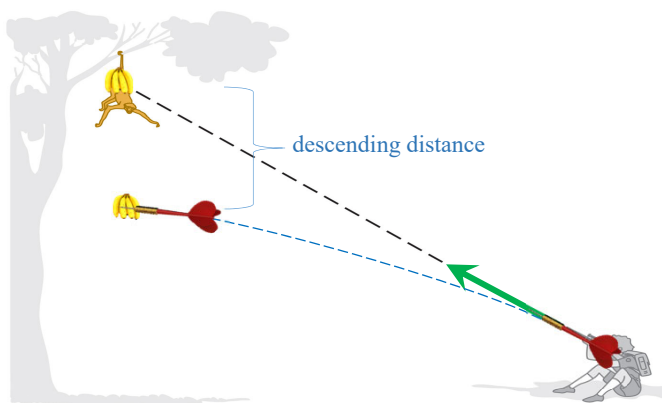


10



11

Problem. A monkey escapes from the zoo with a bunch of bananas and climbs a tree. After failing to entice the monkey down, the zoo keeper fires a tranquilizer dart directly at the monkey. Thinking that the zoo keeper is hungry, the monkey lets go of the bananas at the instant the dart leaves the gun but still refuses to come down from the tree. Will the dart hit the monkey, or banana or nothing? Assume that the effect of air resistance is negligible and both the monkey and the bunch of bananas are treated as point mass. Provide **qualitative** answer.



Answer:

If the gravitational acceleration $g = 0$, the dart will travel on a straight line and hit the monkey (and the bunch of bananas). As both the dart and bananas fall under the same gravitational acceleration and with the same start time, the vertical descending distance for both objects are the same at all times. So **the dart will hit the falling banana** provided the bunch of bananas does not hit the ground first.

12

Projectile Problem in Vector Component Form. In the vector component notation, a particle is projected with a velocity of $(v_0 \cos \theta, v_0 \sin \theta)$ at time $t = 0$ where $45^\circ < \theta < 90^\circ$. The acceleration due to gravity is $(0, -g)$. Find the position vector of the particle when its velocity is at right angle to the initial velocity v_0 .

Answer:

2 vectors (x_1, y_1) and (x_2, y_2) are **perpendicular** to each other if $(x_1, y_1) \cdot (x_2, y_2) = 0$ i.e., $x_1 x_2 + y_1 y_2 = 0$



$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

If $\theta = 90^\circ$, $\cos \theta = 0$
so $\vec{A} \cdot \vec{B} = 0$.

In vector form at time t , the velocity of the projectile is $v_t = (v_0 \cos \theta, v_0 \sin \theta - gt)$.

$$v_t \cdot v_0 = 0 \rightarrow (v_0 \cos \theta, v_0 \sin \theta - gt) \cdot (v_0 \cos \theta, v_0 \sin \theta) = 0$$

$$\rightarrow v_0^2 \cos^2 \theta + v_0^2 \sin^2 \theta - gt v_0 \sin \theta = 0 \rightarrow v_0^2 (\cos^2 \theta + \sin^2 \theta) = gt v_0 \sin \theta \rightarrow t = \frac{v_0}{g \sin \theta}$$

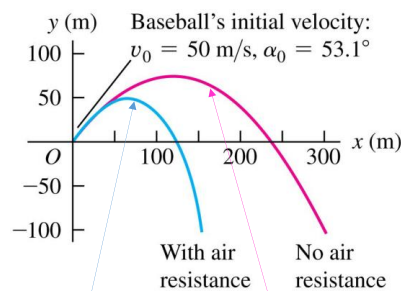
The **position vector** at $t = \frac{v_0}{g \sin \theta}$ is given by the components

$$\begin{aligned} \left(v_0 \cos \theta \times t, v_0 \sin \theta \times t - \frac{1}{2} g t^2 \right) &= \left(v_0 \cos \theta \times \left(\frac{v_0}{g \sin \theta} \right), v_0 \sin \theta \times \left(\frac{v_0}{g \sin \theta} \right) - \frac{1}{2} g \left(\frac{v_0}{g \sin \theta} \right)^2 \right) \\ &= \left(\frac{v_0^2 \cot \theta}{g}, \frac{v_0^2}{2g} \left(\frac{2 \sin^2 \theta - 1}{\sin^2 \theta} \right) \right) = \left(\frac{v_0^2 \cot \theta}{g}, -\frac{v_0^2}{2g} \left(\frac{1 - 2 \sin^2 \theta}{\sin^2 \theta} \right) \right) = \left(\frac{v_0^2 \cot \theta}{g}, -\frac{v_0^2 \cos 2\theta \csc^2 \theta}{2g} \right) \end{aligned}$$

13

The Effects of Air Resistance on Projectile Motion

- Calculations become more complicated.
- Acceleration is not constant.
- Effects can be very large.
- Maximum height and range decrease.
- **Trajectory is no longer a parabola (not symmetrical anymore).**



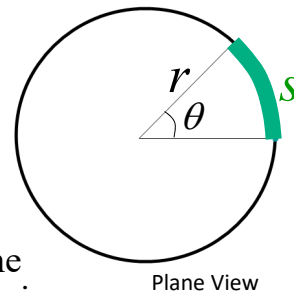
not symmetrical
Tail-end will be asymptotically vertical as the horizontal velocity is approaching 0.

symmetrical

14

Angular Position and Angular Speed

Radian is the measurement of the amount of rotation. We define the angular displacement θ from a reference line as $\theta = \frac{s}{r}$ where s is the arc length and r is the radius, and θ will be in radian.



The positive x -axis is usually used as the reference line for the angular position. In general, counter-clockwise is treated as positive, and clockwise is treated as negative, but these can be re-defined.

The angular speed ω for the rotation is defined as the time rate change of the angular position, i.e., $\omega = \frac{d\theta}{dt}$.

If ω is a constant, the object will be in **uniform** circular motion.

15

Example 2. Uniform circular motion

$$x(t) = r_0 \cos(\theta_0 + \omega_0 t)$$

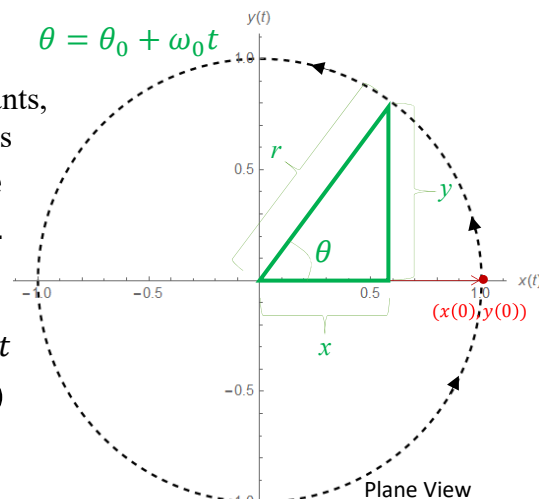
$$y(t) = r_0 \sin(\theta_0 + \omega_0 t)$$

where $r_0 > 0$, and θ_0 and ω_0 are some constants, and at time $t = 0$ the **position** of the particle is $(x(0), y(0)) = (r_0 \cos \theta_0, r_0 \sin \theta_0)$, and the particle is moving on a circle of radius $r = r_0$.

$$\begin{aligned} r^2 &= x^2 + y^2 \\ &= r_0^2 \cos^2(\theta_0 + \omega_0 t) + r_0^2 \sin^2(\theta_0 + \omega_0 t) \\ &= r_0^2 (\cos^2(\theta_0 + \omega_0 t) + \sin^2(\theta_0 + \omega_0 t)) \\ &= r_0^2 \end{aligned}$$

If $r_0 = 1$ m and $\theta_0 = 0^\circ$, then at time $t = 0$,

$$(x(0), y(0)) = r_0(1, 0).$$



In this example the angular speed ω_0 is a constant.

16

Example 2. Uniform circular motion (cont.)

$$\begin{aligned}x(t) &= r_0 \cos(\theta_0 + \omega_0 t) \\y(t) &= r_0 \sin(\theta_0 + \omega_0 t)\end{aligned}$$

The **velocity** of the particle,

$$\begin{aligned}v_x(t) &= \frac{d}{dt}(x(t)) = -\omega_0 r_0 \sin(\theta_0 + \omega_0 t) \\v_y(t) &= \frac{d}{dt}(y(t)) = \omega_0 r_0 \cos(\theta_0 + \omega_0 t)\end{aligned}$$

At time $t = 0$, the **velocity** of the particle is $(v_x(0), v_y(0)) = (-\omega_0 r_0 \sin \theta_0, \omega_0 r_0 \cos \theta_0)$.

The particle is moving with constant **speed** $v = r_0 \omega_0$:

$$v^2 = v_x^2 + v_y^2 = r_0^2 \omega_0^2 \sin^2(\theta_0 + \omega_0 t) + r_0^2 \omega_0^2 \cos^2(\theta_0 + \omega_0 t) = r_0^2 \omega_0^2.$$

Since the particle moves with constant **speed** $v = r_0 \omega_0$ round the circle of radius $r = r_0$, it takes a fixed time T to complete one round. The **period** of the uniform circular motion,

$$\text{velocity} = \frac{\text{distance}}{\text{time}} \rightarrow \text{time} = \frac{\text{distance}}{\text{velocity}}$$

$$T = \frac{2\pi r_0 / r_0}{v / r_0} = \frac{2\pi}{\omega_0}, \text{ or } \omega_0 = \frac{2\pi}{T}.$$

17

Example 2. Uniform circular motion (cont.)

$$\begin{aligned}x(t) &= r_0 \cos(\theta_0 + \omega_0 t) \\y(t) &= r_0 \sin(\theta_0 + \omega_0 t)\end{aligned}$$

If $r_0 = 1$ m, $\theta_0 = 0$, then at time $t = 0$,

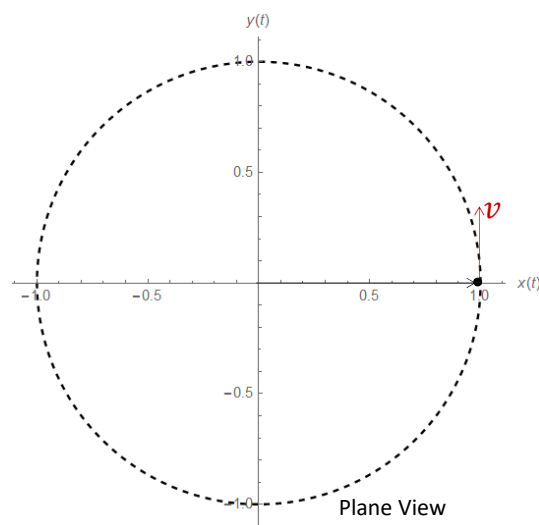
$$\begin{aligned}(x(0), y(0)) &= r_0(\cos \theta_0, \sin \theta_0) \\&= r_0(1, 0).\end{aligned}$$

and,

$$\begin{aligned}(v_x(0), v_y(0)) &= r_0 \omega_0 (-\sin \theta_0, \cos \theta_0) \\&= r_0 \omega_0 (0, 1).\end{aligned}$$

What are the **position** and **velocity** of the particle at time $t = T$?

period



18

Example 2. Uniform circular motion (cont.)

$$\begin{aligned}x(t) &= r_0 \cos(\theta_0 + \omega_0 t) \\y(t) &= r_0 \sin(\theta_0 + \omega_0 t)\end{aligned}$$

$$\begin{aligned}v_x(t) &= -\omega_0 r_0 \sin(\theta_0 + \omega_0 t) \\v_y(t) &= \omega_0 r_0 \cos(\theta_0 + \omega_0 t)\end{aligned}$$

The **acceleration** of the particle is

$$a_x(t) = \frac{d}{dt}(v_x(t)) = -\omega_0^2 r_0 \cos(\theta_0 + \omega_0 t) = -\omega_0^2 x(t)$$

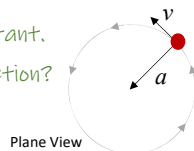
$$a_y(t) = \frac{d}{dt}(v_y(t)) = -\omega_0^2 r_0 \sin(\theta_0 + \omega_0 t) = -\omega_0^2 y(t)$$

The magnitude of **acceleration** for the particle:

$$a^2 = a_x^2 + a_y^2 = \omega_0^4 r_0^2 \cos^2(\theta_0 + \omega_0 t) + \omega_0^4 r_0^2 \sin^2(\theta_0 + \omega_0 t) = \omega_0^4 r_0^2$$

$$\Rightarrow a = \omega_0^2 r_0$$

Magnitude is a constant.
How about the direction?



From $v = r_0 \omega_0$, $\omega_0 = \frac{v}{r_0}$.

$$\text{So } a = \left(\frac{v}{r_0}\right)^2 r_0 = \frac{v^2}{r_0} \quad \left\{ \begin{array}{l} \text{Centripetal} \\ \text{acceleration} \end{array} \right.$$

19

Example 2. Uniform circular motion (cont.)

If $r_0 = 1$ m, $\theta_0 = 0$, then at time $t = 0$,

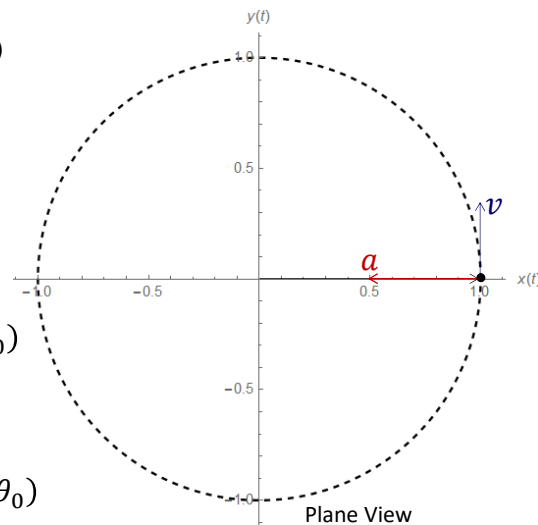
$$\begin{aligned}(x(0), y(0)) &= r_0(\cos \theta_0, \sin \theta_0) \\ &= r_0(1, 0).\end{aligned}$$

and,

$$\begin{aligned}(v_x(0), v_y(0)) &= \omega_0 r_0(-\sin \theta_0, \cos \theta_0) \\ &= \omega_0 r_0(0, 1).\end{aligned}$$

and,

$$\begin{aligned}(a_x(0), a_y(0)) &= \omega_0^2 r_0(-\cos \theta_0, -\sin \theta_0) \\ &= \omega_0^2 r_0(-1, 0).\end{aligned}$$



20

Example 2. Uniform circular motion (cont.)

$$\begin{aligned} x(t) &= r_0 \cos(\theta_0 + \omega_0 t) & v_x(t) &= -\omega_0 r_0 \sin(\theta_0 + \omega_0 t) & a_x(t) &= -\omega_0^2 x(t) \\ y(t) &= r_0 \sin(\theta_0 + \omega_0 t) & v_y(t) &= \omega_0 r_0 \cos(\theta_0 + \omega_0 t) & a_y(t) &= -\omega_0^2 y(t) \end{aligned}$$

If $r_0 = 1$ m, $\theta_0 = 0$, then at time $t = T/8$,

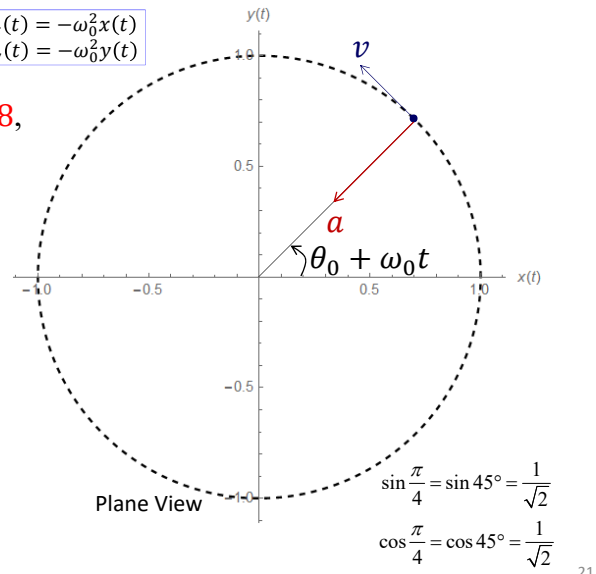
$$\theta_0 + \omega_0 t = 0 + \left(\frac{2\pi}{T}\right) \times \frac{T}{8} = \frac{\pi}{4}.$$

So,

$$(x, y) = r_0 \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right),$$

$$(v_x, v_y) = \omega_0 r_0 \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right),$$

$$(a_x, a_y) = \omega_0^2 r_0 \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right).$$



Example 2. Uniform circular motion (cont.)

In summary, for a particle moving on a circle of radius r_0 , centred at $(0, 0)$, with constant speed $v = \omega_0 r_0$, its **position**,

$$(x, y) = r_0 (\cos(\theta_0 + \omega_0 t), \sin(\theta_0 + \omega_0 t)).$$

Its **velocity**,

$$(v_x, v_y) = \omega_0 r_0 (-\sin(\theta_0 + \omega_0 t), \cos(\theta_0 + \omega_0 t)).$$

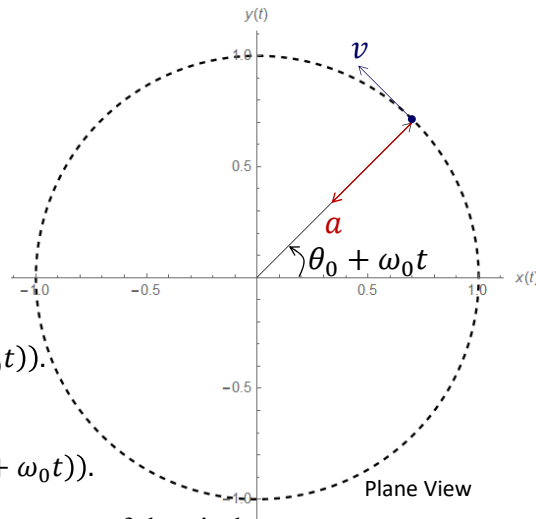
Its **acceleration**,

$$(a_x, a_y) = \omega_0^2 r_0 (-\cos(\theta_0 + \omega_0 t), -\sin(\theta_0 + \omega_0 t)).$$

Its **acceleration** is always directed towards the centre of the circle.

This acceleration is called **centripetal acceleration**,

$$a_c = \omega_0^2 r_0 = \frac{v^2}{r_0} \text{ and it is always perpendicular to its velocity.}$$



Non-uniform

Example 2. Uniform circular motion (cont.)

Remark 2 – Non-uniform circular motion.

In this case, the **speed**

$$v = \omega r_0$$

of the particle is not a constant.

The angular speed ω here is not a constant.

So, in addition to the

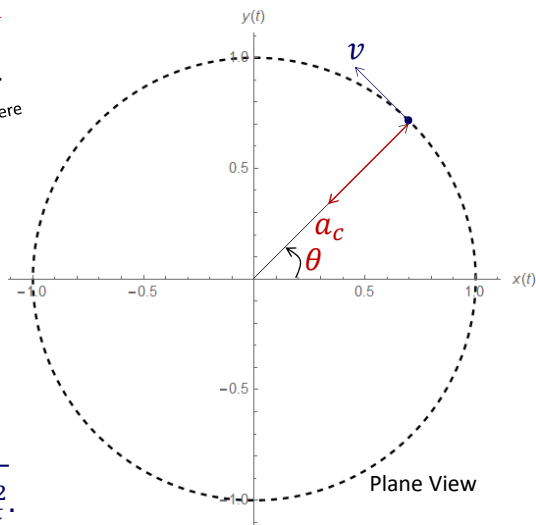
centripetal acceleration $a_c = \omega^2 r_0 = \frac{v^2}{r_0}$,

we also have the **tangential acceleration**

$$a_t = \frac{dv}{dt} = \frac{d\omega}{dt} r_0.$$

The magnitude of the **acceleration** of the particle is $a = \sqrt{a_c^2 + a_t^2}$.

$\frac{d\omega}{dt}$ is referred to as the **angular acceleration**.

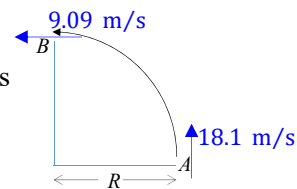


In this example the angular speed is a variable so the tangential speed is also a variable.

23

Problem. Your car moving at 18.1 m/s enters a curve on campus at point A that describes a quarter turn of radius $R = 120$ m from A to B as shown.

You gently apply the brakes, slowing the car down at a constant rate of 0.65 m/s^2 . Just before emerging from the turn at B, what is the magnitude of the net acceleration of your car and the angle between the **net acceleration vector** and the **direction of motion**?



Answer:

The **direction of motion** of the car just **before emerging from the turn at B** is along the **horizontal and to the left**. The net acceleration will be the vector sum of **tangential acceleration** a_t and the **centripetal acceleration** a_c :

$$a_t = -0.65 \text{ m/s}^2, \text{ and } a_c = \frac{v^2}{r} = \frac{v^2}{120} \text{ where } v \text{ is the exit velocity at point B.}$$

The motion of the car along the curve is effectively a 1-dimensional motion with constant acceleration. From $v_f^2 = v_i^2 + 2a\Delta x \rightarrow v_f^2 = (18.1)^2 + 2(-0.65)s \dots (1)$

$$\theta = \frac{s}{r} \quad \left| \quad s = r\theta = 120 \times \frac{\pi}{2} = 60\pi \right| \quad (1) \text{ becomes } v_f^2 = (18.1)^2 - 2(0.65)(60\pi) \rightarrow v_f = 9.09 \text{ m/s} \quad (9.0866)$$

3.14159

24

By the Pythagorean Theorem,
the net acceleration =

$$\begin{aligned}\sqrt{a_c + a_t} &= \sqrt{\left(\frac{v^2}{120}\right)^2 + 0.65^2} \\ &= \sqrt{\left(\frac{9.0866^2}{120}\right)^2 + 0.65^2} \\ &= 0.95 \text{ m/s}^2\end{aligned}$$

The angle between the net acceleration vector and the direction of motion is given by

$$\begin{aligned}180^\circ - \theta &= 180^\circ - \tan^{-1} \frac{|a_c|}{|a_t|} = 180^\circ - \tan^{-1} \left(\frac{\frac{v_f^2}{120}}{0.65} \right) \\ &= 180^\circ - \tan^{-1} \left(\frac{\frac{9.0866^2}{120}}{0.65} \right) = 180^\circ - 46.63^\circ = 133.37^\circ\end{aligned}$$

Problem. Your car moving at 18.1 m/s enters a curve on campus at point A that describes a quarter turn of radius $R = 120$ m from A to B as shown.

You gently apply the brakes, slowing the car down at a constant rate of 0.65 m/s^2 . Just before emerging from the turn at B, what is the magnitude of the net acceleration of your car and the angle between the **net acceleration vector** and the **direction of motion**?

Answer:

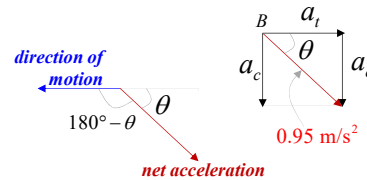
The **direction of motion** of the car just **before emerging from the turn at B** is along the **horizontal and to the left**. The net acceleration will be the vector sum of **tangential acceleration a_t** and the **centripetal acceleration a_c** :

$$a_t = -0.65 \text{ m/s}^2, \text{ and } a_c = \frac{v^2}{r} = \frac{v^2}{120} \text{ where } v \text{ is the exit velocity at point B.}$$

The motion of the car along the curve is effectively a 1-dimensional motion with constant acceleration. From $v_f^2 = v_i^2 + 2a\Delta x \rightarrow v_f^2 = (18.1)^2 + 2(-0.65)s \dots (1)$

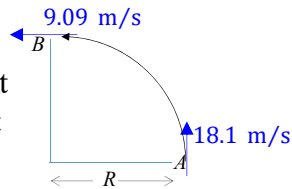
$$\theta = \frac{s}{r} \quad \left| \quad s = r\theta = 120 \times \frac{\pi}{2} = 60\pi \right| \quad (1) \text{ becomes } v_f^2 = (18.1)^2 - 2(0.65)(60\pi) \rightarrow v_f = 9.09 \text{ m/s} \quad (9.0866)$$

3.14159

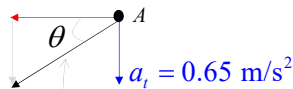


25

What is the net acceleration at point A?



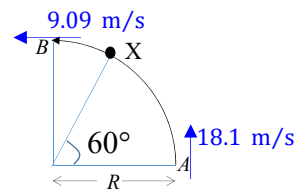
$$a_c = \frac{v^2}{r} = \frac{18.1^2}{120} = 2.73 \text{ m/s}^2$$



$$\text{Net acceleration} = \sqrt{a_c + a_t} = \sqrt{2.73^2 + 0.65^2} = 2.81 \text{ m/s}^2$$

$$\theta = \tan^{-1} \frac{0.65}{2.73} = 13.39^\circ$$

What is the net acceleration at point X?



$$\begin{aligned}v_f^2 &= v_i^2 + 2a\Delta x \rightarrow v_f^2 = (18.1)^2 + 2(-0.65)(120 \times \frac{\pi}{3}) \\ &\rightarrow v_f = 12.82 \text{ m/s}\end{aligned}$$

$$a_c = \frac{v^2}{r} = \frac{12.82^2}{120} = 1.369 \text{ m/s}^2$$

$$\begin{aligned}\text{Net acceleration} &= \sqrt{a_c + a_t} \\ &= \sqrt{1.369^2 + 0.65^2} \\ &= 1.52 \text{ m/s}^2\end{aligned}$$

$$\beta = \tan^{-1} \frac{0.65}{1.369} = 25.40^\circ$$

$$\begin{aligned}\theta &= 30^\circ + (90^\circ - \beta) \\ &= 94.6^\circ\end{aligned}$$

26

Problem. A ball swings counter-clockwise in a vertical circle at the end of a rope 1.50 m long. When the ball is 36.9° on the way up from the x -axis as shown on figure, its (total) acceleration has these components: $a_x = -22.5 \text{ m/s}^2$, and $a_y = -20.2 \text{ m/s}^2$, where x and y are the axes in the Cartesian coordinate system. For that instant,

(i) determine the magnitude of its centripetal acceleration and tangential acceleration

Answer:

Radial component of a_x is $a_x \cos 36.9^\circ = 22.5 \cos 36.9^\circ = 17.9 \text{ m/s}^2$

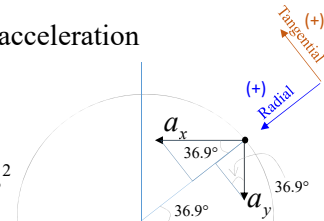
Tangential component of a_x is $a_x \sin 36.9^\circ = 22.5 \sin 36.9^\circ = 13.5 \text{ m/s}^2$

Radial component of a_y is $a_y \sin 36.9^\circ = 20.5 \sin 36.9^\circ = 12.13 \text{ m/s}^2$

Tangential component of a_y is $a_y \cos 36.9^\circ = 20.5 \cos 36.9^\circ = 16.15 \text{ m/s}^2$

Total radial (centripetal) acceleration: $a_x \cos 36.9^\circ + a_y \sin 36.9^\circ = 17.9 + 12.13 = \boxed{29.32 \text{ m/s}^2}$

Total tangential acceleration: $a_x \sin 36.9^\circ - a_y \cos 36.9^\circ = 13.5 - 16.15 = \boxed{-2.65 \text{ m/s}^2}$



27

Total radial (centripetal) acceleration:

$$a_x \cos 36.9^\circ + a_y \sin 36.9^\circ = 17.9 + 12.13 = \boxed{29.32 \text{ m/s}^2}$$

(ii) determine the speed and velocity components of the ball.

Answer:

Centripetal acceleration is given by

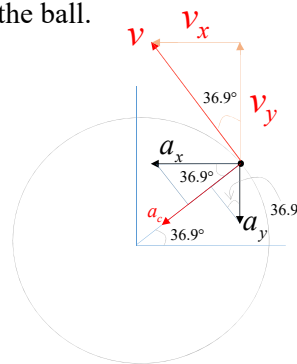
$$a_c = \frac{v^2}{r_0} \rightarrow 29.32 = \frac{v^2}{1.5}$$

$$v^2 = 43.98$$

$$\rightarrow v = \boxed{6.63 \text{ m/s}} \text{ (speed)}$$

x -component of velocity is $v_x = -v \sin 36.9^\circ = -6.63 \sin 36.9^\circ = \boxed{-3.98 \text{ m/s}}$

y -component of velocity is $v_y = v \cos 36.9^\circ = 6.63 \cos 36.9^\circ = \boxed{5.3 \text{ m/s}}$



28

Relative Velocity in One Dimension

Consider this scenario that you are on the waiting platform of Bouna Vista MRT station, and I am inside the train. Take going forward as positive.

- If I am walking forward at 2 m/s relative to the stationary train, and you are not moving, what will be my velocity relative to you (seen by you)? (2 m/s)
- If I am walking forward at 2 m/s in the train, and the train is moving forward at 15 m/s relative to you, what will be my velocity relative to you (seen by you)? ($2+15 = 17$ m/s)
- If I am walking forward at 2 m/s in the stationary train, and you are also walking forward at 2 m/s on the platform, what will be my velocity relative to you? ($2-2 = 0$ m/s → to you I was not moving)
- If I am walking forward at 2 m/s in the stationary train, and you are running forward at 6 m/s on the platform, what will be my velocity relative to you? ($2-6 = -4$ m/s → to you I was moving backward)

29

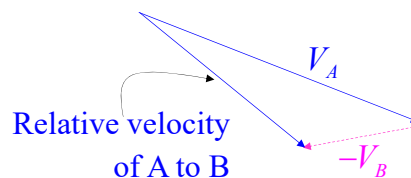
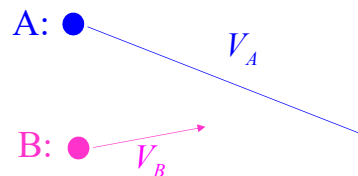
Relative Velocity in Vector Form

A: ● ————— 5 m/s →

B: ● ——— 1.8 m/s →

The relative velocity

of A to B is $V_A - V_B = 5 - 1.8$
 $= 3.2$ m/s
 (observed by B,
 or seen by B)



30

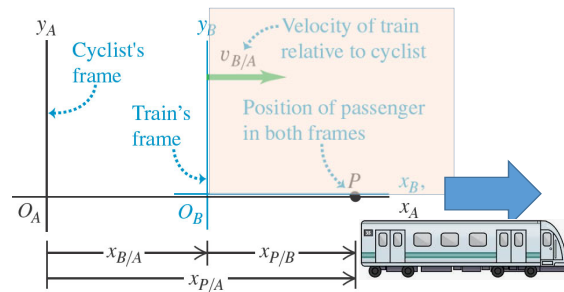
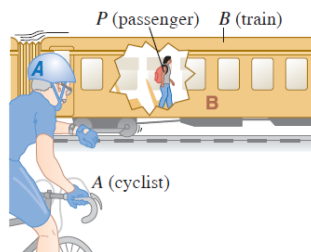
Relative Velocity in One Dimension

- If point P is moving relative to reference frame A , we denote the velocity of P relative to frame A as $v_{P/A}$.

- If P is moving relative to frame B and frame B is moving relative to frame A all along the same direction, then the velocity of P relative to frame A is

Galilean velocity transformation

$v_{P/A} = v_{P/B} + v_{B/A}$. For example, if I am walking forward at 2 m/s relative to the MRT train, and the train is moving forward at 15 m/s relative to you, my velocity relative to you will be $2 + 15 = 17$ m/s.



31

Galilean Velocity Transformation



32

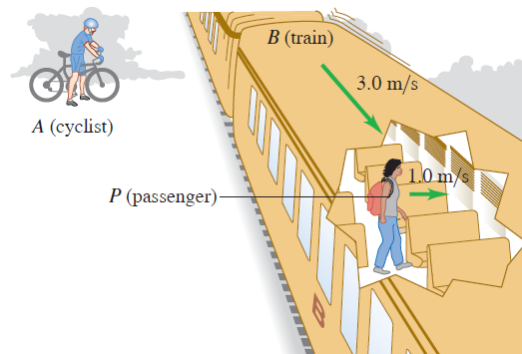
Galilean Velocity Transformation



33

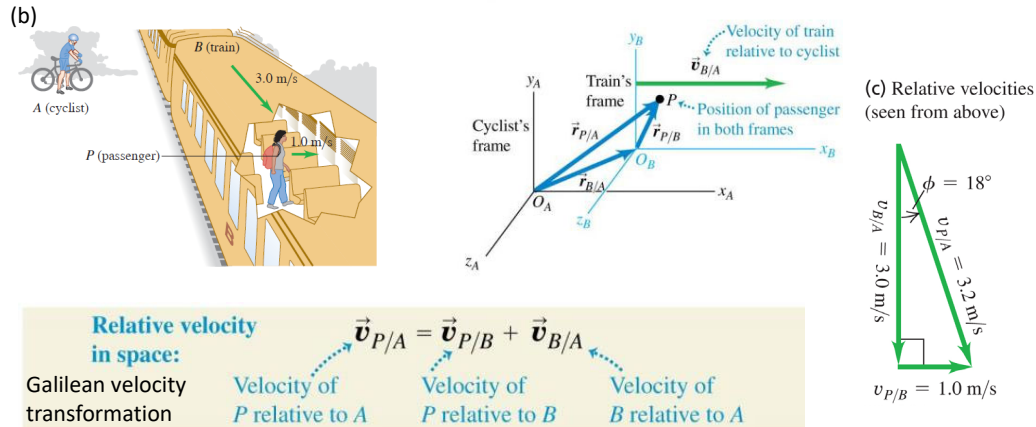
Relative Velocity in Two or Three Dimensions

- We extend relative velocity to **two or three dimensions** by using vector addition to combine velocities.



34

Relative Velocity in Two or Three Dimensions



Vector addition is commutative, i.e., $\vec{A} + \vec{B} = \vec{B} + \vec{A}$.

35

Example. An airplane's compass indicates that it is headed due north, and its air-speed indicator shows that it is moving through the air at 240 km/h. If there is a 100 km/h wind from west to east, what is the velocity of the airplane relative to the earth?

Answer:

By the Galilean velocity transformation, $\vec{v}_{\text{Airplane/Earth}} = \vec{v}_{\text{Airplane/Wind}} + \vec{v}_{\text{Wind/Earth}}$

$$|\vec{v}_{\text{Airplane/E}}| = \sqrt{240^2 + 100^2} = 260 \text{ km/h (Pythagorean Theorem)}$$

$$\alpha = \tan^{-1}\left(\frac{100}{240}\right) = 22.62^\circ \text{E of N}$$

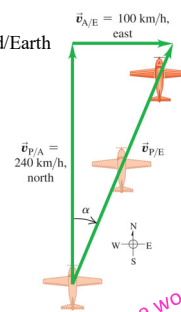
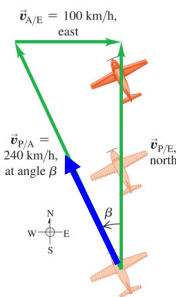
The crosswind increases the speed of the airplane relative to the Earth, but pushes the airplane off course. So in what direction should pilot head to travel due north?

Answer:

$$\beta = \sin^{-1}\left(\frac{100}{240}\right) = 24.62^\circ \text{W of N, and the ground speed will be}$$

$\sqrt{240^2 - 100^2} = 218.17 \text{ km/h.}$ Not only the headwind, the crosswind can also reduce the airplane's speed relative to the Earth. This is an unfortunate fact of aeronautical life.

Is $\beta = \alpha$? No !!!



Please work on your Tutorial-1.
No marking, but solutions will be presented in class.

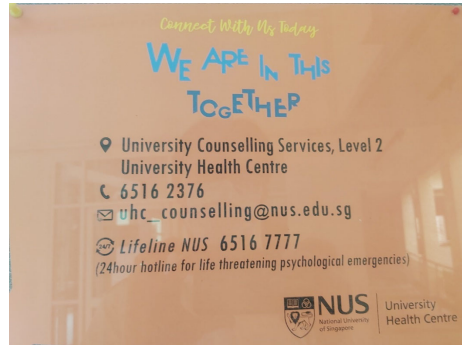
36

Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019